

The monotone relaxation of Levine’s hat problem has success probability $\frac{1}{2}$ for every number of players

A disproof of Conjecture 2.2 of Alon, Friedgut, Kalai and Kindler

github.com/graph-theory-AI

Machine-generated note

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Abstract

In Levine’s hat problem, t players each wear a stack of n hats, independently coloured black or white with probability $\frac{1}{2}$; each player sees every stack but her own and must point to a black hat on her own head, and the players win only if all succeed. Alon, Friedgut, Kalai and Kindler introduced the relaxation in which a player may, instead of a single hat, name any *balanced monotone* family of colourings of her own stack, and conjectured (Conjecture 2.2 of arXiv:2208.06858) that the optimal success probability $p_{\text{monotone}}(t)$ of this relaxed game tends to 0 as $t \rightarrow \infty$. We show that $p_{\text{monotone}}(t) = \frac{1}{2}$ for every $t \geq 1$. The construction partitions each stack into N tribes of m hats; the players aim at the event that some tribe is entirely black across all players, and each player rounds the conditional fibre of that event to a balanced monotone family. A second-moment bound shows the conditional probabilities concentrate around $\frac{1}{2}$, so the loss in the rounding is $O(t 2^{-m/2})$.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 5](#) records the full provenance.

1 Introduction

The game and its relaxations. Fix integers $t \geq 1$ (players) and $n \geq 1$ (hats per player). Write $\Omega_n = \{0, 1\}^n$ with the uniform probability measure μ ; a point $x_i \in \Omega_n$ is the colouring of player i ’s stack (1 = black), and $x = (x_1, \dots, x_t) \in \Omega_n^t$ is uniformly distributed. For a set $\mathcal{W}$ of “admissible” subsets of Ω_n , a *strategy* is a t -tuple of functions $f_i : \Omega_n^{t-1} \rightarrow \mathcal{W}$, and its *winning set* is

$$W = \{x \in \Omega_n^t : x_i \in f_i(x_{-i}) \text{ for all } i\},$$

where x_{-i} denotes the stacks of all players other than i [1, Section 2]. The value of the game is $p(t, n) = \max \mu(W)$ over all strategies and $p(t) = \lim_{n \rightarrow \infty} p(t, n)$. Three choices of $\mathcal{W}$ are considered in [1]: dictator families (“point to hat j ”), giving Levine’s original game and p_{dict} ; intersecting families, giving $p_{\text{intersecting}}$; and *balanced monotone families*, giving p_{monotone} . Here a family $\mathcal{A} \subseteq \Omega_n$ is *monotone* if $x \in \mathcal{A}$ and $y \geq x$ coordinatewise imply $y \in \mathcal{A}$, and *balanced* if $|\mathcal{A}| = 2^{n-1}$. Since a dictator family is intersecting and an intersecting family of size 2^{n-1} is balanced monotone,

$$p_{\text{monotone}}(t, n) \geq p_{\text{intersecting}}(t, n) \geq p_{\text{dict}}(t, n).$$

Levine’s conjecture (Conjecture 1.1 of [1]) is $p_{\text{dict}}(t) \rightarrow 0$; [1] proves that $p_{\text{dict}}(t)$ is strictly decreasing in t , and states two progressively stronger conjectures.

Conjecture 1.1 ([1, Conjecture 2.1]). $p_{\text{intersecting}}(t)$ tends to 0 as t grows.

Conjecture 1.2 ([1, Conjecture 2.2]). $p_{\text{monotone}}(t)$ tends to 0 as t grows.

Throughout, $\mathcal{M}_n$ denotes the set of balanced monotone families in Ω_n , so a monotone strategy is a tuple $f_i : \Omega_n^{t-1} \rightarrow \mathcal{M}_n$; equivalently, $W \subseteq \Omega_n^t$ is the winning set of a monotone strategy iff every i -fibre $\{x_i : (x_1, \dots, x_t) \in W\}$, for every fixed x_{-i} , is contained in some member of $\mathcal{M}_n$ (and W can then be enlarged to the set where equality holds).

Result.

Theorem 1.3 (Main theorem). *For every integer $t \geq 1$,*

$$p_{\text{monotone}}(t) = \frac{1}{2}.$$

More precisely, $p_{\text{monotone}}(t, n) \leq \frac{1}{2}$ for all n , and for every $t \geq 2$ and $m \geq 1$ there are n and a monotone strategy with n hats per player whose success probability is at least

$$\frac{1}{2} - \sqrt{2} t 2^{-m/2} - \frac{t}{2} 2^{-mt}.$$

Corollary 1.4. *Conjecture 1.2 is false. Conjecture 1.1 and Levine’s Conjecture 1.1 are not affected.*

Proof. $p_{\text{monotone}}(t) = \frac{1}{2} \not\equiv 0$. The balanced monotone families used in the proof of Theorem 1.3 are in general not intersecting (Remark 4.2), so nothing follows for $p_{\text{intersecting}}$ or p_{dict} . $\square$

Idea of proof. The upper bound is trivial: whatever player i sees, she names a set of measure exactly $\frac{1}{2}$ for her own independent stack, so she alone succeeds with probability $\frac{1}{2}$. For the lower bound, split each stack into N tribes of m hats and let C be the monotone event “some tribe is all-black for all t players”; N is chosen so that $\Pr(C) \approx \frac{1}{2}$. Given x_{-i} , the i -fibre of C is again monotone, with conditional measure $Q_i = 1 - (1 - 2^{-m})^{M_i}$ where $M_i \sim \text{Bin}(N, 2^{-m(t-1)})$ counts the tribes that are all-black for the other players. A product-variance bound gives $\text{Var}(Q_i) \leq 2 \cdot 2^{-m}$, so Q_i concentrates at its mean $\Pr(C)$. Each player rounds her fibre to a nested balanced monotone family (Lemma 2.1); by a union bound the rounding loses at most $\sum_i \mathbb{E}|Q_i - \frac{1}{2}| = O(t 2^{-m/2})$ of success probability (Lemma 2.2). Letting $m \rightarrow \infty$ gives $p_{\text{monotone}}(t) \geq \frac{1}{2}$. [full proof]

2 Two general lemmas

Lemma 2.1 (Balancing a monotone family). *Let $\mathcal{F} \subseteq \Omega_n$ be monotone. There is a balanced monotone family $\mathcal{A} \in \mathcal{M}_n$ with $\mathcal{F} \subseteq \mathcal{A}$ if $|\mathcal{F}| \leq 2^{n-1}$ and $\mathcal{A} \subseteq \mathcal{F}$ if $|\mathcal{F}| \geq 2^{n-1}$; in either case $\mu(\mathcal{F} \triangle \mathcal{A}) = |\mu(\mathcal{F}) - \frac{1}{2}|$. With a fixed tie-breaking order on Ω_n the construction is deterministic, i.e. $\mathcal{A}$ is a function of $\mathcal{F}$.*

Idea of proof. If $\mathcal{F}$ is too small, repeatedly add an inclusion-maximal element of the complement; if too large, repeatedly remove an inclusion-minimal element of $\mathcal{F}$. Both operations preserve monotonicity and change the size by one. [full proof]

Lemma 2.2 (Synchronisation). *Let $C \subseteq \Omega_n^t$ be monotone (as a subset of $\{0, 1\}^{nt}$). For each player i and each x_{-i} let $\mathcal{F}_i(x_{-i}) = \{x_i : (x_1, \dots, x_t) \in C\}$ be the i -fibre of C , $q_i(x_{-i}) = \mu(\mathcal{F}_i(x_{-i}))$, and let $\mathcal{A}_i(x_{-i}) \in \mathcal{M}_n$ be the balanced monotone family obtained from $\mathcal{F}_i(x_{-i})$ by Lemma 2.1. Then $f_i = \mathcal{A}_i$ is a monotone strategy, and its winning set $E = \bigcap_i E_i$, $E_i = \{x : x_i \in \mathcal{A}_i(x_{-i})\}$, satisfies*

$$\Pr(E) \geq \Pr(C) - \sum_{i=1}^t \mathbb{E} \left| q_i(X_{-i}) - \frac{1}{2} \right|. \quad (1)$$

Idea of proof. Each fibre of a monotone set is monotone, so Lemma 2.1 applies. Conditionally on x_{-i} , the set $C \setminus E_i$ has measure $(q_i - \frac{1}{2})_+$ by nestedness; then $\Pr(E) \geq \Pr(C) - \sum_i \Pr(C \setminus E_i)$. [full proof]

3 The block-tribes construction

Fix $t \geq 2$ and $m \geq 1$, and put

$$u = 2^{-m}, \quad s = u^t = 2^{-mt}, \quad N = \left\lceil \frac{\log 2}{-\log(1-s)} \right\rceil, \quad n = mN. \quad (2)$$

Each player's n hats are partitioned into N tribes of m hats; write $X_{i,a,b} \in \{0, 1\}$ for the colour of hat $b \in [m]$ of tribe $a \in [N]$ of player $i \in [t]$. Let

$$C = \{\exists a \in [N] \forall i \in [t] \forall b \in [m] : X_{i,a,b} = 1\}$$

be the event that some tribe is entirely black across all players. C is monotone.

Lemma 3.1 (Properties of C). *With the notation of (2) and $c := \Pr(C)$:*

- (a) $c = 1 - (1-s)^N$ and $\frac{1}{2} \leq c < \frac{1}{2} + \frac{s}{2}$.
- (b) For each player i , conditionally on X_{-i} , the i -fibre of C has measure $Q_i = 1 - (1-u)^{M_i}$, where $M_i \sim \text{Bin}(N, r)$ with $r = u^{t-1}$ is the number of tribes whose $m(t-1)$ hats of the other players are all black. Moreover $\mathbb{E}Q_i = c$.
- (c) $\text{Var}(Q_i) \leq Nsu \leq 2u$.
- (d) $\mathbb{E}|Q_i - \frac{1}{2}| \leq \sqrt{2u} + \frac{s}{2}$.

Idea of proof. (a) The N tribe events are independent of probability s ; the two inequalities are the definition of N as a ceiling. (b) Given the others' hats, C happens iff one of the M_i eligible tribes is all-black for player i ; $\mathbb{E}Q_i = 1 - (1-ru)^N$ and $ru = s$. (c) $1 - Q_i$ is a product of N independent factors in $\{1, 1-u\}$; the variance of a product of independent $[0, 1]$ -valued variables is at most the sum of the variances, which is $Nr(1-r)u^2 \leq Nsu$, and $Ns \leq \log 2 + s < 2$ by the choice of N . (d) Cauchy–Schwarz plus (a). [\[full proof\]](#)

Lemma 3.2 (Limit versus supremum). $p_{\text{monotone}}(t, n)$ is nondecreasing in n and at most $\frac{1}{2}$; hence $p_{\text{monotone}}(t) = \lim_n p_{\text{monotone}}(t, n) = \sup_n p_{\text{monotone}}(t, n)$.

Idea of proof. Append an ignored hat: $\mathcal{A} \mapsto \mathcal{A} \times \{0, 1\}$ maps $\mathcal{M}_n$ into $\mathcal{M}_{n+1}$ and preserves the winning set's measure. The upper bound is the trivial one-player bound. [\[full proof\]](#)

4 Remarks

Remark 4.1 (Size of the stacks). The construction uses $n = mN$ hats with $N \approx (\log 2)/s = (\log 2) 2^{mt}$, so n is enormous, and doubly exponential in t if m grows with t (for instance $m = \lceil 6 \log_2(t+1) \rceil$ makes the bound of [Theorem 1.3](#) equal to $\frac{1}{2} - O(t^{-2})$). This is legitimate: $p_{\text{monotone}}(t)$ is defined as the limit in n for fixed t , and it is consistent with the remark in [\[1\]](#) that $p(t, n) \rightarrow 0$ as $t \rightarrow \infty$ for each fixed n .

Remark 4.2 (Scope). The families produced by [Lemma 2.1](#) from the fibres of C are balanced and monotone but in general not intersecting: two members of the completed family may have disjoint supports. Hence the argument says nothing about $p_{\text{intersecting}}$ or p_{dict} ; Conjectures 1.1 and 2.1 of [\[1\]](#) remain open.

Remark 4.3 (Computational checks). The referee ([Appendix B](#)) verified [Lemma 2.1](#) exhaustively on all 168 monotone families of $\{0, 1\}^4$; verified the whole construction for $t = 2, m = 1$ ($N = 3, 64$ outcomes) in exact rational arithmetic, obtaining $\Pr(C) = 37/64$, $\Pr(E_1) = \Pr(E_2) = \frac{1}{2}$ exactly and success probability $19/64$ against the bound $11/64$ from [\(1\)](#); and evaluated $c, \mathbb{E}Q_i, \text{Var}(Q_i)$ and $\mathbb{E}|Q_i - \frac{1}{2}|$ exactly for $t \in \{2, 3, 4, 10\}$ and $m \leq 20$, where the rigorous lower bound $c - t \mathbb{E}|Q_i - \frac{1}{2}|$ on the success probability reaches 0.4994 for $t = 2$ and 0.4968 for $t = 10$ at $m = 20$.

Remark 4.4 (Novelty and related work). The referee’s search found no earlier resolution of [Conjecture 1.2](#). Heilman and Tamuz [2] give a Fourier-analytic upper bound for two players in the original game and do not treat the monotone relaxation. A recent analysis of a continuous-stack relaxation [3] finds value exactly $\frac{1}{2}$ for that different relaxation, an independent echo of the phenomenon here. None of this has been checked by a human.

5 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The construction and proof were found by the language model gpt-5.6-so1 in a single autonomous pass on 2026-08-31, as part of a large sweep over open problems catalogued from arXiv (catalog id 2208.06858__01; original writeup in attacks/2208.06858__01/output.md). (2) An independent adversarial referee agent (claude-fable-5) re-derived every step, checked the definitions against the source paper, verified the balancing lemma exhaustively on all 168 monotone families of $\{0, 1\}^4$, verified the whole construction end-to-end in exact rational arithmetic for $t = 2$ and one or two hats per tribe, and evaluated the concentration bounds exactly for tribe counts up to $N \sim 10^{60}$; its verdict was CONFIRMED. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-09 the writeup was rewritten into the present note by claude-fable-5.1: the text was reorganised with full proofs in [Appendix A](#), and one clarification supplied by the referee was added as [Lemma 3.2](#) (the source paper defines $p_{\text{monotone}}(t)$ as a limit in the stack length n , the writeup argued with a supremum; the two agree because $p_{\text{monotone}}(t, n)$ is nondecreasing in n). No other mathematical content was changed. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

A Full proofs

Lemma 2.1 (Balancing a monotone family). *Let $\mathcal{F} \subseteq \Omega_n$ be monotone. There is a balanced monotone family $\mathcal{A} \in \mathcal{M}_n$ with $\mathcal{F} \subseteq \mathcal{A}$ if $|\mathcal{F}| \leq 2^{n-1}$ and $\mathcal{A} \subseteq \mathcal{F}$ if $|\mathcal{F}| \geq 2^{n-1}$; in either case $\mu(\mathcal{F} \triangle \mathcal{A}) = |\mu(\mathcal{F}) - \frac{1}{2}|$. With a fixed tie-breaking order on Ω_n the construction is deterministic, i.e. $\mathcal{A}$ is a function of $\mathcal{F}$.*

Proof. Suppose $|\mathcal{F}| < 2^{n-1}$. The complement $\Omega_n \setminus \mathcal{F}$ is nonempty and downward closed (if $y \notin \mathcal{F}$ and $x \leq y$ then $x \notin \mathcal{F}$, since $x \in \mathcal{F}$ would force $y \in \mathcal{F}$). Let z be an inclusion-maximal element of $\Omega_n \setminus \mathcal{F}$ (the first such in the tie-breaking order) and put $\mathcal{F}' = \mathcal{F} \cup \{z\}$. Then $\mathcal{F}'$ is monotone: if $x \in \mathcal{F}'$ and $y \geq x$, either $x \in \mathcal{F}$ and $y \in \mathcal{F}$, or $x = z$ and then $y \geq z$ with $y \neq z$ gives $y \in \mathcal{F}$ by maximality of z in the complement, while $y = z \in \mathcal{F}'$. Repeating $2^{n-1} - |\mathcal{F}|$ times yields a monotone $\mathcal{A} \supseteq \mathcal{F}$ with $|\mathcal{A}| = 2^{n-1}$, and $\mu(\mathcal{F} \triangle \mathcal{A}) = \mu(\mathcal{A}) - \mu(\mathcal{F}) = \frac{1}{2} - \mu(\mathcal{F})$.

Suppose $|\mathcal{F}| > 2^{n-1}$. Let z be an inclusion-minimal element of $\mathcal{F}$ and put $\mathcal{F}' = \mathcal{F} \setminus \{z\}$. Then $\mathcal{F}'$ is monotone: if $x \in \mathcal{F}'$ and $y \geq x$ then $y \in \mathcal{F}$, and $y = z$ would give $z \geq x$ with $x \in \mathcal{F}$, $x \neq z$, contradicting minimality of z . Repeating $|\mathcal{F}| - 2^{n-1}$ times yields a monotone $\mathcal{A} \subseteq \mathcal{F}$ with $|\mathcal{A}| = 2^{n-1}$ and $\mu(\mathcal{F} \triangle \mathcal{A}) = \mu(\mathcal{F}) - \frac{1}{2}$. If $|\mathcal{F}| = 2^{n-1}$ take $\mathcal{A} = \mathcal{F}$. Note 2^{n-1} is an integer for $n \geq 1$, so exact balance is attainable. Choosing at each step the first admissible element in the fixed order makes $\mathcal{A}$ a function of $\mathcal{F}$. $\square$

Lemma 2.2 (Synchronisation). *Let $C \subseteq \Omega_n^t$ be monotone (as a subset of $\{0, 1\}^{nt}$). For each player i and each x_{-i} let $\mathcal{F}_i(x_{-i}) = \{x_i : (x_1, \dots, x_t) \in C\}$ be the i -fibre of C , $q_i(x_{-i}) = \mu(\mathcal{F}_i(x_{-i}))$, and let $\mathcal{A}_i(x_{-i}) \in \mathcal{M}_n$ be the balanced monotone family obtained from $\mathcal{F}_i(x_{-i})$ by [Lemma 2.1](#). Then $f_i = \mathcal{A}_i$ is a monotone strategy, and its winning set $E = \bigcap_i E_i$, $E_i = \{x : x_i \in \mathcal{A}_i(x_{-i})\}$, satisfies*

$$\Pr(E) \geq \Pr(C) - \sum_{i=1}^t \mathbb{E} \left| q_i(X_{-i}) - \frac{1}{2} \right|. \quad (1)$$

Proof. If $(x_1, \dots, x_t) \in C$ and $x'_i \geq x_i$, then $(x_1, \dots, x'_i, \dots, x_t) \geq (x_1, \dots, x_t)$ coordinatewise, so it lies in C ; hence each fibre $\mathcal{F}_i(x_{-i})$ is monotone and Lemma 2.1 applies, producing $\mathcal{A}_i(x_{-i}) \in \mathcal{M}_n$ as a function of x_{-i} . So $(f_i)_i = (\mathcal{A}_i)_i$ is a monotone strategy with winning set $E = \bigcap_i E_i$.

Fix i and condition on x_{-i} ; then x_i is uniform on Ω_n . The set $\{x_i : x \in C \setminus E_i\} = \mathcal{F}_i(x_{-i}) \setminus \mathcal{A}_i(x_{-i})$. If $q_i(x_{-i}) \leq \frac{1}{2}$ then $\mathcal{F}_i \subseteq \mathcal{A}_i$ and this set is empty; if $q_i(x_{-i}) > \frac{1}{2}$ then $\mathcal{A}_i \subseteq \mathcal{F}_i$ and it has measure $\mu(\mathcal{F}_i) - \mu(\mathcal{A}_i) = q_i(x_{-i}) - \frac{1}{2}$. Thus

$$\Pr(C \setminus E_i \mid x_{-i}) = \left(q_i(x_{-i}) - \frac{1}{2} \right)_+ \leq \left| q_i(x_{-i}) - \frac{1}{2} \right|,$$

and taking expectations, $\Pr(C \setminus E_i) \leq \mathbb{E}|q_i(X_{-i}) - \frac{1}{2}|$. Finally, $C \setminus E \subseteq \bigcup_i (C \setminus E_i)$, so by the union bound

$$\Pr(E) \geq \Pr(C \cap E) = \Pr(C) - \Pr(C \setminus E) \geq \Pr(C) - \sum_{i=1}^t \Pr(C \setminus E_i),$$

which is (1). $\square$

Lemma 3.1 (Properties of C). *With the notation of (2) and $c := \Pr(C)$:*

- (a) $c = 1 - (1-s)^N$ and $\frac{1}{2} \leq c < \frac{1}{2} + \frac{s}{2}$.
- (b) For each player i , conditionally on X_{-i} , the i -fibre of C has measure $Q_i = 1 - (1-u)^{M_i}$, where $M_i \sim \text{Bin}(N, r)$ with $r = u^{t-1}$ is the number of tribes whose $m(t-1)$ hats of the other players are all black. Moreover $\mathbb{E}Q_i = c$.
- (c) $\text{Var}(Q_i) \leq Nsu \leq 2u$.
- (d) $\mathbb{E}|Q_i - \frac{1}{2}| \leq \sqrt{2u} + \frac{s}{2}$.

Proof. (a) For $a \in [N]$ let T_a be the event that all mt hats of tribe a are black. The T_a depend on disjoint sets of hats, so they are independent, each of probability $2^{-mt} = s$, and $C = \bigcup_a T_a$ gives $c = 1 - (1-s)^N$. Put $\lambda = \log 2 / (-\log(1-s)) > 0$, so $N = \lceil \lambda \rceil$. From $N \geq \lambda$, $(1-s)^N \leq (1-s)^\lambda = \frac{1}{2}$, i.e. $c \geq \frac{1}{2}$. From $N < \lambda + 1$, $(1-s)^N > (1-s)^{\lambda+1} = \frac{1}{2}(1-s)$, i.e. $c < \frac{1}{2} + \frac{s}{2}$.

(b) Fix i and condition on X_{-i} . Call tribe a *eligible* if $X_{j,a,b} = 1$ for all $j \neq i$ and all $b \in [m]$; eligibility of the N tribes is determined by disjoint sets of $m(t-1)$ hats, so the number M_i of eligible tribes is $\text{Bin}(N, r)$ with $r = 2^{-m(t-1)} = u^{t-1}$. Given X_{-i} , the event C holds iff for some eligible tribe a all of player i 's hats in tribe a are black; the i -fibre of C is thus the set of $x_i \in \Omega_n$ having some eligible tribe all-black, a monotone family whose measure is $Q_i = 1 - (1-u)^{M_i}$ (the M_i eligible tribes involve disjoint sets of m hats of player i , each all-black with probability u). By independence of the tribes, $\mathbb{E}Q_i = 1 - \mathbb{E}[(1-u)^{M_i}] = 1 - \prod_{a=1}^N \mathbb{E}[Y_a]$ with $Y_a = 1-u$ if tribe a is eligible and $Y_a = 1$ otherwise, so $\mathbb{E}Y_a = 1 - ru$ and $\mathbb{E}Q_i = 1 - (1 - ru)^N = 1 - (1-s)^N = c$, using $ru = u^t = s$.

(c) With Y_a as in (b), $Z_i := 1 - Q_i = \prod_{a=1}^N Y_a$ with $Y_1, \dots, Y_N$ independent and $0 \leq Y_a \leq 1$. For independent random variables $Y_a \in [0, 1]$,

$$\text{Var}\left(\prod_{a=1}^N Y_a\right) = \prod_{a=1}^N \mathbb{E}[Y_a^2] - \prod_{a=1}^N (\mathbb{E}Y_a)^2 = \sum_{a=1}^N \left(\prod_{b < a} (\mathbb{E}Y_b)^2 \right) (\mathbb{E}[Y_a^2] - (\mathbb{E}Y_a)^2) \left(\prod_{b > a} \mathbb{E}[Y_b^2] \right) \leq \sum_{a=1}^N \text{Var}(Y_a),$$

since every factor in the two products is in $[0, 1]$. Here Y_a takes the value $1-u$ with probability r and 1 with probability $1-r$, so $\text{Var}(Y_a) = r(1-r)u^2 \leq ru^2 = su$. Hence $\text{Var}(Q_i) = \text{Var}(Z_i) \leq Nsu$. Finally, $-\log(1-s) \geq s$ gives $\lambda \leq \log 2/s$, so $Ns < (\lambda + 1)s \leq \log 2 + s < 2$ (as $s \leq \frac{1}{4}$), and $\text{Var}(Q_i) \leq 2u$.

(d) By the triangle inequality, Cauchy–Schwarz, (b), (c) and (a),

$$\mathbb{E}\left|Q_i - \frac{1}{2}\right| \leq \mathbb{E}|Q_i - c| + \left|c - \frac{1}{2}\right| \leq \sqrt{\text{Var}(Q_i)} + \left|c - \frac{1}{2}\right| \leq \sqrt{2u} + \frac{s}{2}. \quad \square$$

Lemma 3.2 (Limit versus supremum). $p_{\text{monotone}}(t, n)$ is nondecreasing in n and at most $\frac{1}{2}$; hence $p_{\text{monotone}}(t) = \lim_n p_{\text{monotone}}(t, n) = \sup_n p_{\text{monotone}}(t, n)$.

Proof. Upper bound. Let (f_i) be any monotone strategy with n hats and winning set $W \subseteq E_1 = \{x : x_1 \in f_1(x_{-1})\}$. Since x_1 is independent of x_{-1} and $\mu(f_1(x_{-1})) = \frac{1}{2}$ for every x_{-1} , $\Pr(E_1) = \mathbb{E}_{x_{-1}}[\mu(f_1(x_{-1}))] = \frac{1}{2}$, so $\mu(W) \leq \frac{1}{2}$ and $p_{\text{monotone}}(t, n) \leq \frac{1}{2}$.

Monotonicity in n . Given a monotone strategy (f_i) with n hats, define a strategy with $n + 1$ hats by $f'_i(x'_{-i}) = f_i(x_{-i}) \times \{0, 1\}$, where x_{-i} is obtained from x'_{-i} by deleting the last hat of every stack. If $\mathcal{A} \in \mathcal{M}_n$ then $\mathcal{A} \times \{0, 1\}$ is monotone and has $2 \cdot 2^{n-1} = 2^n$ elements, so it lies in $\mathcal{M}_{n+1}$. The winning set of (f'_i) is $W \times \{0, 1\}^t$ (the last hats are ignored), of measure $\mu(W)$. Hence $p_{\text{monotone}}(t, n + 1) \geq p_{\text{monotone}}(t, n)$. A bounded nondecreasing sequence converges to its supremum. $\square$

Theorem 1.3 (Main theorem). For every integer $t \geq 1$,

$$p_{\text{monotone}}(t) = \frac{1}{2}.$$

More precisely, $p_{\text{monotone}}(t, n) \leq \frac{1}{2}$ for all n , and for every $t \geq 2$ and $m \geq 1$ there are n and a monotone strategy with n hats per player whose success probability is at least

$$\frac{1}{2} - \sqrt{2}t 2^{-m/2} - \frac{t}{2} 2^{-mt}.$$

Proof. The upper bound $p_{\text{monotone}}(t, n) \leq \frac{1}{2}$ for all n is Lemma 3.2, and gives $p_{\text{monotone}}(t) \leq \frac{1}{2}$. For $t = 1$ any balanced family wins with probability exactly $\frac{1}{2}$, so $p_{\text{monotone}}(1) = \frac{1}{2}$.

Let $t \geq 2$ and $m \geq 1$, and take u, s, N, n and the event C as in Section 3. Apply Lemma 2.2 to C : for each player i and each x_{-i} , the fibre $\mathcal{F}_i(x_{-i})$ has measure $q_i(x_{-i}) = Q_i$ in the notation of Lemma 3.1(b), and $\mathcal{A}_i(x_{-i})$ is its balanced monotone completion (enlarged if $Q_i \leq \frac{1}{2}$, shrunk if $Q_i > \frac{1}{2}$). This is a deterministic monotone strategy with $n = mN$ hats per player, and by (1) and Lemma 3.1(a),(d) its success probability satisfies

$$\Pr(E) \geq c - \sum_{i=1}^t \mathbb{E} \left| Q_i - \frac{1}{2} \right| \geq \frac{1}{2} - t \left(\sqrt{2}u + \frac{s}{2} \right) = \frac{1}{2} - \sqrt{2}t 2^{-m/2} - \frac{t}{2} 2^{-mt}.$$

This is the quantitative statement. Hence $p_{\text{monotone}}(t, mN) \geq \frac{1}{2} - \sqrt{2}t 2^{-m/2} - \frac{t}{2} 2^{-mt}$, and by Lemma 3.2 $p_{\text{monotone}}(t) = \sup_n p_{\text{monotone}}(t, n) \geq \frac{1}{2} - \sqrt{2}t 2^{-m/2} - \frac{t}{2} 2^{-mt}$ for every m . Letting $m \rightarrow \infty$ gives $p_{\text{monotone}}(t) \geq \frac{1}{2}$, and with the upper bound, $p_{\text{monotone}}(t) = \frac{1}{2}$. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-fable-5, 2026-09-02) on the original writeup. Its step numbers refer to that writeup, not to this note. The scripts it mentions are in `verification/scripts/2208.06858__01/`. Treat it as machine output, not as an authority.

Target: Conjecture 2.2 of Alon–Friedgut–Kalai–Kindler, arXiv:2208.06858 (“The success probability in Levine’s hat problem, and independent sets in graphs”): $p_{\text{monotone}}(t) \rightarrow 0$ as $t \rightarrow \infty$.

Writeup’s claim: **disproved** — a block-tribes construction gives balanced monotone strategies with success probability arbitrarily close to $1/2$ for every fixed t , so $p_{\text{monotone}}(t) = 1/2$ for all t .

B.1 Interpretation

I fetched the source paper (arXiv:2208.06858v2, PDF read in full for Sections 1–2) and compared the writeup’s formalization against it verbatim.

The paper (Section 2, pp. 3–4) defines: a strategy for the t -player game is a t -tuple $(f_1, \dots, f_t)$ with $f_i : B^{t-1} \rightarrow \mathcal{W}$; the winning set of the strategy is $\{x : \forall i, x_i \in f_i(x_{-i})\}$; $p(t, n) := \max_{W \in \mathcal{W}^{(t)}} \mu(W)$ and $p(t) := \lim_{n \rightarrow \infty} p(t, n)$. For the monotone relaxation, $\mathcal{W}$ is “the set of all balanced monotone families in $\{0, 1\}^n$, i.e. all W containing precisely half the points in $\{0, 1\}^n$ such that if $x \in W$ and $y_i \geq x_i$ for all i then $y \in W$.” Conjecture 2.2 is exactly “ $p_{\text{monotone}}(t)$ tends to 0 as t grows.”

The writeup’s formalization — player i , after seeing x_{-i} , selects an arbitrary balanced upward-closed family $\mathcal{A}_i(x_{-i})$, succeeding when $x_i \in \mathcal{A}_i(x_{-i})$; equivalently every i -fiber of the common success set is contained in a balanced monotone family — **matches the paper’s definition exactly**. This is not a strawman reading: the family is allowed to depend arbitrarily on the observed stacks in the paper’s own definition, and Conjecture 2.2 is stated for precisely this quantity. A reasonable author of the source paper would consider Conjecture 2.2 resolved (negatively) by this argument. The writeup correctly notes it leaves Conjectures 2.1 ($p_{\text{intersecting}}$) and 1.1 (p_{dict} , Levine’s original conjecture) untouched, and it does not contradict anything proven in the source paper (Theorem 1.2 concerns p_{dict} ; the paper’s remark that $p(t, n) \rightarrow 0$ for *fixed* n is consistent, since the construction takes $n \approx m(\log 2)2^{mt}$ growing with t).

One convention nuance: the paper defines $p(t)$ as $\lim_n p(t, n)$, the writeup speaks of the supremum over stack lengths. These coincide: $p_{\text{monotone}}(t, n)$ is nondecreasing in n (extend any balanced monotone $W \subseteq \{0, 1\}^n$ to $W \times \{0, 1\} \subseteq \{0, 1\}^{n+1}$, which is again balanced and monotone, and let strategies ignore the extra bit), and it is bounded above by $1/2$ for every n (see step 2 below), so the limit exists and equals the supremum. This one-line observation is not spelled out in the writeup but is immediate; I verified it.

B.2 Step-by-step findings

step	label	note
1. Formalization of p_{monotone} (Statement section)	VALID	Matches arXiv:2208.06858 Section 2 verbatim (fiber condition, balanced = exactly 2^{n-1} points, upward closed, max over strategies, limit in n).
2. Upper bound $p_{\text{monotone}}(t) \leq 1/2$	VALID	For any strategy, $\Pr(x_i \in \mathcal{A}_i(x_{-i})) =$ $\mathbb{E}_{x_{-i}}[\mu(\mathcal{A}_i(x_{-i}))] = 1/2$ since x_i is independent of x_{-i} and every $\mathcal{A}_i$ is balanced; the winning set is contained in each E_i . Holds for every n , so also in the limit.

step	label	note
3. Lemma 1 (balancing a monotone family)	VALID	Adding an inclusion-maximal element of the (downward-closed) complement preserves upward closure, as does removing an inclusion-minimal element; one element at a time reaches exactly 2^{n-1} ; nestedness gives $\mu(\mathcal{F} \triangle \mathcal{A}) = \mu(\mathcal{F}) - 1/2 $. Verified exhaustively by code on all 168 monotone families of $\{0, 1\}^4$ (see Computational check).
4. Synchronization inequality (1)	VALID	With the enlarge-if-small / shrink-if-large rule, $\mu(C \setminus E_i \mid x_{-i}) = (q_i(x_{-i}) - 1/2)_+ \leq q_i - 1/2 $, and $\Pr(\cap_i E_i) \geq \Pr(C) - \sum_i \Pr(C \setminus E_i)$ by union bound. Re-derived; also verified exactly on exhaustive toy instances.
5. Block-tribes event C , eq. (2)	VALID	C is monotone; the N tribe events are independent with probability $s = u^t$ each, so $c = 1 - (1 - s)^N$. With $N = \lceil \log 2 / (-\log(1 - s)) \rceil$: $N \geq \log 2 / (-\log(1 - s))$ gives $c \geq 1/2$; $N < \log 2 / (-\log(1 - s)) + 1$ gives $c < \frac{1}{2} + \frac{s}{2}$ (indeed $c < \frac{1}{2}(1 + s)$ via $(1 - s)^N > \frac{1}{2}(1 - s)$). Re-derived and confirmed numerically for all tested (t, m) .
6. Fiber structure, eqs. (3)–(4)	VALID	Given x_{-i} , a tribe is eligible iff its $m(t - 1)$ exposed bits are all black (probability $r = u^{t-1}$, independent across tribes), so $M_i \sim \text{Bin}(N, r)$; the fiber is “some eligible tribe all-black for player i ”, measure $Q_i = 1 - (1 - u)^{M_i}$, monotone. $\mathbb{E}Q_i = 1 - (1 - ru)^N = 1 - (1 - s)^N = c$ since $ru = s$. Verified exactly (both exhaustively on toys and to machine precision in the binomial computation).

step	label	note
7. Variance bound, eq. (5)	VALID	$Z_i = \prod_{a=1}^N Y_a$ with independent $Y_a \in \{1, 1-u\}$, $\text{Var}(Y_a) = r(1-r)u^2$, $\mathbb{E}[Y_a^2] \leq 1$; telescoping gives $\text{Var}(\prod Y_a) \leq \sum_a \text{Var}(Y_a) = Nr(1-r)u^2 \leq Nsu$ (as $ru^2 = su$). $Ns \leq \log 2 + s < 2$ from the definition of N and $-\log(1-s) \geq s$. Hence $\text{Var}(Q_i) \leq 2u$. Re-derived in full; numerically the true variance is far below $2u$ in every tested case.
8. L^1 concentration, eq. (6)	VALID	$\mathbb{E} Q_i - 1/2 \leq \mathbb{E} Q_i - c + c - 1/2 \leq \sqrt{\text{Var}(Q_i)} + s/2$ by Cauchy–Schwarz and (2). Confirmed numerically with wide margins.
9. Final bound, eq. (7), and $p_{\text{monotone}}(t) \geq 1/2$	VALID	Combining (1),(2),(6): success $\geq c - t(\sqrt{2u} + s/2) \geq \frac{1}{2} - \sqrt{2}t2^{-m/2} - \frac{t}{2}2^{-mt} \rightarrow \frac{1}{2}$ as $m \rightarrow \infty$ for fixed t . The strategy is deterministic and valid (each $\mathcal{A}_i(x_{-i})$ balanced monotone, a function of x_{-i} only). Arithmetic checked ($\sqrt{2u} = \sqrt{2}2^{-m/2}$, $s = 2^{-mt}$).
10. Conclusion $p_{\text{monotone}}(t) = 1/2$ for every t	VALID	Follows from steps 2 and 9 plus monotonicity of $p_{\text{monotone}}(t, n)$ in n (one-line cylinder extension, not stated in the writeup but immediate; verified — see Interpretation). $t = 1$ is trivially $1/2$.
11. Explicit rate remark ($m = \lceil 6 \log_2(t+1) \rceil$, $n \approx m(\log 2)2^{mt}$)	VALID	$\sqrt{2}t(t+1)^{-3} = O(t^{-2})$; $N \approx (\log 2)/s = (\log 2)2^{mt}$, $n = mN$. Non-load-bearing remark; checked.
12. Scope (does not resolve intersecting/dict variants)	VALID	Correct: the balanced completions of tribes-fibers are not intersecting (two members can have disjoint supports), so nothing follows for $p_{\text{intersecting}}$ or p_{dict} ; Conjectures 2.1 and 1.1 and Levine’s conjecture remain open.

No step is a GAP or ERROR. The single unstated detail (limit vs. supremum over n , step 10) is a routine one-line observation that I verified myself.

B.3 Reference check

- **arXiv:2208.06858 (v2, Aug 2023), Alon–Friedgut–Kalai–Kindler.** Fetched the PDF and read Sections 1–2 in full. Confirmed: the strategy/winning-set definitions (t -tuple of functions $f_i : B^{t-1} \rightarrow \mathcal{W}$, fiber condition), $p(t, n) = \max_W \mu(W)$, $p(t) = \lim_n p(t, n)$, the definition of balanced monotone families, the chain $p_{\text{monotone}}(t, n) \geq p_{\text{intersecting}}(t, n) \geq p_{\text{dict}}(t, n)$, and the exact statement of Conjecture 2.2 (“ $p_{\text{monotone}}(t)$ tends to 0 as t grows”). All exactly as the writeup uses them. The paper contains no lower-bound discussion for p_{monotone} , so the construction contradicts nothing proven there.
- The writeup is otherwise **self-contained**: it invokes no external theorems (Lemma 1, the union bound, Cauchy–Schwarz, and elementary variance facts are all proved or standard and were re-derived here).
- **Prior-resolution search (is this ALREADY_KNOWN?).** Targeted searches (arXiv, web) for any resolution of Conjecture 2.2 / the monotone relaxation: Heilman–Tamuz “A Fourier approach to Levine’s hat puzzle” (arXiv:2503.09042) is a two-player upper bound via Chang’s lemma and never mentions the monotone relaxation; “An analytical framework for the Levine hats problem” (arXiv:2508.01737, Aug 2025) proves the *continuous-stack* game has value exactly $1/2$ (related in spirit — relaxations of the game degenerate to $1/2$ — but a different relaxation, with no mention of monotone families or Conjecture 2.2); Buhler et al. (arXiv:1407.4711) concerns the dictator game. No paper disproving (or proving) Conjecture 2.2 was found, consistent with the catalog’s May 2026 review. As far as I can determine the result is new.

B.4 Computational check

Scripts (reproducible) in `verification/scripts/2208.06858__01/`:

1. `check_lemma1.py` — enumerates all upward-closed families of $\{0, 1\}^4$ ($168 = \text{Dedekind}(4)$, as expected) and runs the greedy Lemma-1 balancing on each with a fixed tie-break. Result: **168/168 pass** (output balanced, monotone, nested the right way, symmetric difference = $||F| - 8|$; 0 failures).
2. `check_end_to_end.py` — full exhaustive verification of the construction for $t = 2, m = 1$ using the writeup’s own $N = \lceil \log 2 / (-\log(3/4)) \rceil = 3$, i.e. $n = 3$ hats per player, all $2^6 = 64$ outcomes. Verified exactly (rational arithmetic): $\Pr(C) = 37/64 = 1 - (1 - s)^N$; C globally monotone; every fiber monotone with measure $1 - (1 - u)^{M_i}$; $\mathbb{E}Q_i = \Pr(C)$ exactly; every balanced family valid, $\Pr(E_1) = \Pr(E_2) = 1/2$ exactly; exact success probability of the constructed strategy = $19/64 = 0.296875 \geq \text{bound (1)} = 0.171875$; eq. (2) ($1/2 \leq 37/64 < 5/8$), eq. (5) ($\text{Var}(Q) = 0.0662 \leq 2u = 1$) and eq. (6) all hold. A second exhaustive case ($t = 2, m = 2, N = 2$, 256 outcomes) verified the same machinery, including inequality (1), with $\Pr(E_i) = 1/2$ exact.
3. `check_bound_numeric.py` — exact binomial-sum evaluation (stable mode-anchored recurrence; window ± 40 SDs, missing mass $< 10^{-170}$) of $c, \mathbb{E}Q, \text{Var}(Q), \mathbb{E}|Q - 1/2|$ for $t \in \{2, 3, 4, 10\}$, $m \in \{2, \dots, 20\}$ (N up to $\sim 10^{60}$). All of (2), (4), (5), (6) hold in every case, with wide margins (e.g. true $\text{Var}(Q)$ is 1–2 orders of magnitude below the bound $2u$; $\mathbb{E}Q - c$ is 0 to machine precision). The rigorous lower bound $c - t \mathbb{E}|Q - 1/2|$ on the success probability reaches, at $m = 20$: **0.499351** ($t = 2$), **0.499027** ($t = 3$), **0.498703** ($t = 4$), **0.496756** ($t = 10$) — converging to $1/2$ as claimed.

No claimed property failed computationally.

B.5 Caveats

- **Limit vs. supremum over n .** The paper defines $p(t) = \lim_n p(t, n)$; the writeup argues via a supremum. The gap is closed by the standard observation that $p_{\text{monotone}}(t, n)$ is nondecreasing in n (cylinder extension $W \mapsto W \times \{0, 1\}$ preserves balance and monotonicity), which the writeup omits but which is immediate.
- **Enormous stacks.** The construction needs $n \approx m(\log 2)2^{mt}$ hats per player (double-exponential in t if m grows with t). This is entirely legitimate under the paper’s definition (limit over n for each fixed t), and is consistent with the paper’s remark that $p(t, n) \rightarrow 0$ for each fixed n .
- **Scope.** Only Conjecture 2.2 is refuted. Conjecture 2.1 ($p_{\text{intersecting}}(t) \rightarrow 0$) and Levine’s Conjecture 1.1 remain open; the balanced monotone families produced are not intersecting, and the writeup says so explicitly.
- **Edge cases.** $t = 1$: trivially $p_{\text{monotone}}(1) = 1/2$ (any balanced family), consistent with the theorem statement “for every $t \geq 1$ ”. The construction itself is used for $t \geq 2$. Balancing to exactly 2^{n-1} points is always possible since 2^n is even for $n \geq 1$.
- **Sociological caveat.** The refuted conjecture is by Alon, Friedgut, Kalai and Kindler; that such a clean construction was missed is surprising but not implausible — the paper contains no lower-bound analysis of the monotone relaxation, and the degeneration of relaxed variants to $1/2$ has a recent independent echo in the continuous-stack game (arXiv:2508.01737, Theorem 32). This does not affect the mathematical verdict.

B.6 Referee summary

I set out assuming the writeup was wrong and failed to find an error. The formalization of p_{monotone} matches the source paper’s Section 2 definitions verbatim (checked against the fetched PDF), so there is no interpretation gaming: Conjecture 2.2 as the authors stated it is what is refuted. Every step was re-derived: the balancing lemma, the synchronization inequality, the tribes probability computations, the product-variance bound $\text{Var}(Q_i) \leq Nsu \leq 2u$, and the final estimate (7) are all correct; the only unstated detail (monotonicity of $p_{\text{monotone}}(t, n)$ in n , needed to pass from large- n constructions to the limit defining $p_{\text{monotone}}(t)$) is a one-line standard fact I verified. The computational check confirmed Lemma 1 exhaustively on all 168 monotone families of $\{0, 1\}^4$, verified the entire construction end-to-end exhaustively at $t = 2, m = 1$ (exact success 19/64, all identities exact in rational arithmetic), and evaluated the concentration bounds exactly for parameters up to $N \sim 10^{60}$, with the rigorous lower bound on the success probability reaching 0.4994 ($t = 2$) and 0.4968 ($t = 10$) at $m = 20$ and converging to $1/2$. A literature search found no prior resolution of Conjecture 2.2. Verdict: **CONFIRMED** — the writeup correctly disproves Conjecture 2.2 of arXiv:2208.06858, establishing $p_{\text{monotone}}(t) = 1/2$ for every t , while Conjectures 2.1 and 1.1 remain open.

References

- [1] N. Alon, E. Friedgut, G. Kalai and G. Kindler, The success probability in Levine’s hat problem, and independent sets in graphs, arXiv:2208.06858 (v2, 2023).
- [2] S. Heilman and O. Tamuz, A Fourier approach to Levine’s hat puzzle, arXiv:2503.09042 (2025).
- [3] An analytical framework for the Levine hats problem, arXiv:2508.01737 (2025).